Retrospective Study

Ganglion Impar Neurolysis Enhances the Short-term Efficacy of Pudendal Nerve Pulsed Radiofrequency Treatment for Pudendal Neuralgia: A Retrospective Cohort Study

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Background: Pudendal neuralgia (PN) lacks specific treatments and severely affects patients' quality of life. Although pudendal nerve pulsed radiofrequency (PRF) can alleviate symptoms, its overall efficacy is not satisfactory. Ganglion impar neurolysis (GIN) has been widely employed for perineal pain management. We hypothesize that combining PRF with GIN may enhance therapeutic outcomes for PN compared to PRF monotherapy. To the best of our knowledge, no relevant reports have been found.

Objective: To assess whether GIN improves PRF treatment efficacy for PN.

Study Design: A retrospective cohort study.

Setting: This is a single-center retrospective study conducted at the Department of Pain Medicine, Peking University People's Hospital in Beijing, China.

Methods: We conducted a retrospective analysis of 84 women patients with PN treated from January 2020 through February 2024. The patients were divided into 2 groups: PRF Group (n = 40) received only pudendal nerve PRF; PRF + GIN Group (n = 44) received PRF combined with GIN. We recorded the posttreatment differences in Visual Analog Scale (VAS) scores, Overall Improvement in Pain (OIP) values, Hospital Anxiety and Depression Scales scores (anxiety: HADS-A; depression: HADS-D), and Pain Catastrophizing Scale scores (PCS) of the 2 groups at time points up to posttreatment month 6.

Results: At one, 2, 3, and 6 months posttreatment, the VAS scores in both groups were lower than pretreatment levels ($P < 0.05$). At 6 months posttreatment, the mean (SD) VAS score in the PRF+GIN Group (2.09 [1.38]) was lower than that in the PRF Group (3.81 [1.85]) ($P < 0.05$). The mean (SD) OIP value in the PRF+GIN group (73.72% [17.27%]) was higher than that in the PRF Group (56.13% [26.90%]) ($P < 0.05$). At 3 and 6 months posttreatment, the HAD scores and PCS scores in both groups were lower than pretreatment levels ($P < 0.05$). At 6 months posttreatment, the mean (SD) HADS-A and HADS-D scores in the PRF+GIN Group—3.91 (2.70) and 3.37 (2.18), respectively—were lower than those in the PRF group—6.54 (3.75) and 5.55 (3.25), respectively ($P < 0.05$); the mean (SD) PCS score in the PRF+GIN Group 12.16 (6.16) was lower than that in the PRF Group 19.52 (9.92) ($P < 0.05$). No severe complications were observed at any posttreatment time point.

Limitations: This single-center study has a small sample size, a short follow-up period, and lacks patients who are men.

Conclusions: GIN enhances short-term pudendal nerve PRF efficacy for PN, with no significant adverse reactions.

Key words: Pudendal neuralgia, pulsed radiofrequency, ganglion impar neurolysis

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Pudendal neuralgia (PN) is a neuropathic pain disorder affecting the perineal region. Current estimates suggest an incidence of approximately one in100,000, with women predominating(1). Due to diagnostic challenges and delayed clinical presentation, the actual prevalence of PN is likely considerably higher than what is reported. Patients with PN typically experience pain within the nerve's sensory distribution, accompanied by anorectal dysesthesia, urinary urgency/frequency, sexual dysfunction, and psychiatric comorbidities, all severely impairing quality of life (2). With its poorly understood etiology and no definitive treatment, PN remains a complex clinical challenge.

Therapeutic strategies for PN include behavioral modification, pelvic floor physical therapy, oral medications (anticonvulsants, antidepressants, and analgesics), pudendal nerve block, pudendal nerve radiofrequency, pudendal nerve decompression, and sacral nerve stimulation (3-5). Pulsed radiofrequency (PRF), a non-neurodestructive treatment for neuropathic pain, is the preferred treatement for patients who are refractory to conservative therapies, such as pharmacotherapy or physical rehabilitation (6). Recent studies report pudendal nerve PRF can alleviate PN symptoms (7-10). However, its overall efficacy remains unsatisfactory, requiring additional measures for better pain relief.

The ganglion impar, the most caudal ganglion of the sympathetic chain, contains sympathetic nerve fibers originating from the perineal region and partial pelvic organs. Ganglion impar interventions are indicated for coccydynia, chronic prostatitis, chronic proctitis, pelvic malignancies, and chronic perineal pain. Previous studies have demonstrated that ganglion impar neurolysis (GIN) can alleviate chronic pelvic-perineal pain of diverse etiologies (11-13). Based on this rationale, combining GIN may enhance the efficacy of pudendal nerve PRF for treating PN.

Our retrospective analysis of clinical data compared the therapeutic outcomes between pudendal nerve PRF combined with GIN and PRF monotherapy in patients with PN. We aimed to evaluate whether this combination therapy has superior efficacy to PRF monotherapy. If superior efficacy is found, it should inform updating and improving therapeutic algorithms for this challenging condition.

Methods

Patients

We conducted a retrospective analysis of women inpatients with PN who were treated in the Department of Pain Medicine at Peking University People's Hospital from January 2020 through February 2024. This study was approved by the Ethics Committee of Peking University People's Hospital (Approval No. 2024PHB200-001).

Inclusion criteria were:

- 18—80 years old
- A pudendal neuralgia diagnosis meeting the Nantes Criteria (14)
  - pain localized to the anatomical distribution of the pudendal nerve
  - pain exacerbated by sitting
  - pain does not wake the patient at night
  - pain without objective sensory deficits
  - pain is relieved by a diagnostic pudendal nerve block (relief ≥ 50%)
- Patients who received pudendal nerve PRF monotherapy or pudendal nerve PRF combined with GIN
- Completion of a baseline assessment and 6-month posttreatment follow-up.

Exclusion criteria were:

- Pain caused by pelvic malignancy
- Pain caused by pelvic surgery
- Use of adjunctive therapies during follow-up (e.g., traditional Chinese medicine, pudendal nerve acupuncture.

A total of 97 patients with PN were assessed for eligibility for our study; 84 were included in the final analysis. Patients were divided into 2 groups: the PRF Group (n = 40), those who received pudendal nerve PRF monotherapy from January 2020 through November 2022; and the PRF+GIN Group (n = 44), those who received pudendal nerve PRF combined with GIN from December 2022 through February 2024.

Treatment Procedure

All procedures were conducted by a single experienced pain management specialist skilled in chronic pelvic pain management.

Pudendal nerve PRF procedure

Patients were positioned prone. After local anesthesia was administered at the puncture site, a 22G needle (BeiQi Medical Technology Co., Ltd.) was inserted under ultrasound guidance (Resona I9, Mindray Bio-Medical Electronics Co., Ltd.) through the sacrotuberous ligament to the medial side of the pudendal artery (Fig 1). A radiofrequency generator (RFE 4, BeiQi

Medical Technology Co., Ltd.) was then connected. Sensory testing (50 Hz) was performed, and the needle was fine-tuned until a sensory disturbance was elicited in the perineal target area when the voltage was below 0.3 V, indicating that the needle tip had reached the pudendal nerve region. Motor testing (2 Hz) was then conducted. If the voltage exceeded 2 V without inducing any movement in the lower limb on the puncture side, it was considered that the needle tip was not near the sciatic nerve. PRF was then applied with the following parameters: temperature of 42°C, frequency of 2 Hz, pulse width of 20 milliseconds, and duration of 900 seconds. For patients experiencing bilateral pain, bilateral pudendal nerve PRF was performed.

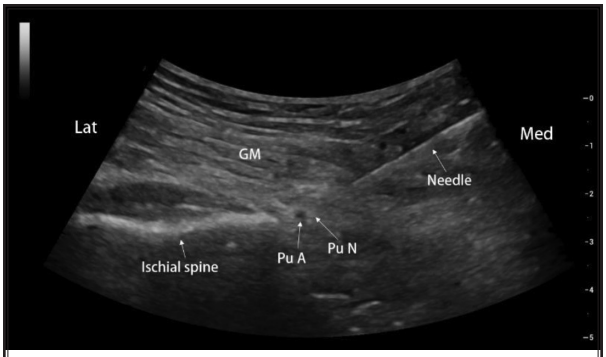


Fig. 1. *The puncture pathway diagram for pudendal nerve pulsed radiofrequency. GM, gluteus maximus; Pu A, pudendal artery; Pu N, pudendal nerve.*

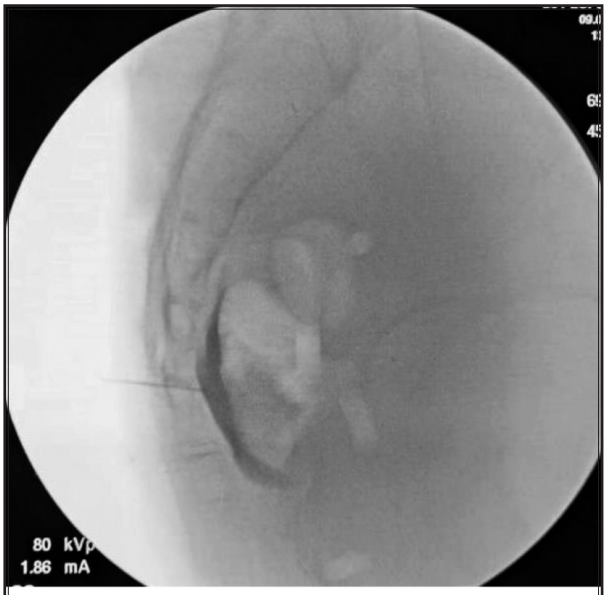


Fig. 2. *Ganglion impar puncture under x-ray guidance (via sacroiliac joint).*

GIN procedure

The patient was positioned prone. After local anesthesia was administered at the puncture site, a 22G needle (BeiQi Medical Technology Co., Ltd.) was inserted via a posterior midline approach through the sacrococcygeal disc to the anterior side of the ventral sacrococcygeal ligament. A loss of resistance was felt, indicating that the needle had reached the target site. After confirming no blood return, one mL of contrast medium was injected; the spread into the anterior sacrococcygeal region was observed in the lateral x-ray view (Fig 2). After confirming accurate needle placement, 1 mL of 2% lidocaine was injected. Following a 5-minute observation period without the patient having any adverse reactions, 5 mL of 95% ethanol was injected.

Data Collection

Pretreatment Data

Patient information, including age, gender, body mass index (BMI [kg/m^2]), pain duration, pain location, pain characteristics and pain intensity, were collected through the inpatient information system of Peking University People's Hospital. Patients with chronic pelvic pain treated at our center routinely complete the hospital anxiety and depression scale (HADS) (HADS-A: anxiety; HADS-D: depression) (15) and the Pain Catastrophizing Scale (PCS) (16).

Posttreatment Data

Posttreatment follow-up data were collected through the chronic pelvic pain follow-up system of the Department of Pain Medicine, Peking University People's Hospital. All data in this system were collected by the same medical assistant, who distributed electronic questionnaires to the patients.

Pain Relief

Pain levels were assessed using the Visual Analog Scale (VAS) at posttreatment months one, 2, 3, and 6 months. The VAS is a 10 centimeter line labeled as 0 is no pain and 10 is the most severe pain imaginable; the patient's score is measured in millimeters. The Overall Improvement in Pain (OIP) score was also recorded. The OIP is the patient's perceived overall improvement in pain (pain intensity, range, frequency, etc.) compared to pretreatment levels, and is scored 0%—100%. A score of 0% indicates no change or worsening of pain,

while 100% represents complete pain relief. An OIP score ≥ 50% is considered efficacious, and ≥ 75% is considered excellent.

Changes in Anxiety, Depression, and Pain Catastrophizing

Changes in anxiety, depression, and pain catastrophizing were assessed using HADS and PCS scores at posttreatment months 3 and 6.

Adverse Reactions and Complications

Local bleeding; infection; nerve damage; lower limb numbness; urinary and defecatory dysfunction; and other serious reactions that occured posttreatment were recorded.

Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics 27.0 (IBM Corporation) Categorical data are presented as frequency (%), and intergroup comparisons were made using the χ^2 test. Continuous data that followed a normal distribution are expressed as mean (SD), with comparisons between the 2 groups made using the independent samples t test. Comparisons of different observational parameters between the 2 groups at various time points were made using repeated measures analysis of variance. A 2-tailed *P* value of < 0.05 was considered statistically significant.

Results

Patient Demographics

A total of 97 women patients with PN were initially assessed for eligibility for our study. Six were excluded due to secondary PN from pelvic malignancies or prior surgery. During follow-up, 3 patients in the PRF Group and 4 in the PRF+GIN Group were excluded after they received electroacupuncture stimulation of the pudendal nerve and Chinese Herbal Medicine, respectively. Ultimately, 84 patients (PRF Group, n = 40; PRF+GIN Group, n = 44) were included in the final analysis (Fig. 3). Baseline characteristics (age; BMI; pain duration, location, and characteristics; and diagnostic pudendal nerve block outcomes) were similar between the groups (*P* > 0.05) (Table 1).

VAS Scores

In the PRF Group, the mean VAS scores decreased from a pretreatment score of 7.00 (1.81) to 3.22 (1.52), 3.09 (2.02), 3.65 (1.91), and 3.81 (1.85) at posttreatment months one,, 2, 3, and 6. In the PRF+GIN Group, the mean pretreatment score of 7.19 (1.71) decreased to 3.31 (1.62), 3.22 (1.92), 3.09 (1.85), and 2.09 (1.38) at the same time points (Fig. 4). There were no significant pretreatment differences between groups. At posttreatment, both groups had significant reductions (*P* < 0.05), with the PRF+GIN Group having a lower VAS score that was statistically significant at 6 months (*P* < 0.05), but no other time points differed in statistical significance.

Overall Improvement in Pain

At posttreatment months one, 2, and 3, the PRF Group had mean OIP values of 63.54% (22.88%) 61.77% (25.25%), and 56.45% (28.26%), respectively. In the PRF+GIN Group, mean OIP values were 68.59% (15.09%), 66.25 (21.92%), and 68.28% (23.09%), at the same time points (Fig. 5). However, at posttreatment month 6, the OIP value was statistically significantly higher in the PRF+GIN Group 73.72% (17.27%) over the PRF Group 56.13% (26.90%), (*P* < 0.05) (Fig. 5). No statistically significant differences were noted at other time points. The excellent rate was also higher in the PRF+GIN Group at posttreatment months 3 and 6 (*P* < 0.05). While the PRF Group had a decline in both efficacy and excellent rate over time, the PRF+GIN Group saw a decrease in efficacy but an increase in excellent rate (Table 2). OIP ≥ 50% is considered effective, and OIP ≥ 75% is considered excellent.

Mental State

Anxiety (HADS-A Scores)

In the PRF Group, the mean pretreatment HADS-A scores decreased from 9.32 (3.35) to 6.55 (3.32) and 6.54 (3.75) at posttreatment months 3 and 6. In the PRF+GIN Group, scores decreased from 9.50 (3.62) at pretreatment to 4.12 (2.57) and 3.91 (2.70) at 3 and 6 months posttreatment, respectively (Fig. 6). Both groups had statistically significant reductions at posttreatment months 3 and 6 (*P* < 0.05); the PRF+GIN Group had statistically significantly lower scores than the PRF Group (*P* < 0.05).

Depression (HADS-D Scores)

In the PRF Group, mean HADS-D scores decreased from 9.64 (3.61) at pretreatment to 6.20 (3.42) and 5.55 (3.25) at posttreatment months 3 and 6. In the PRF+GIN Group, scores decreased from 10.30 (3.83) at pretreatment to 4.22 (2.40) and 3.37 (2.18) at 3 and 6 months

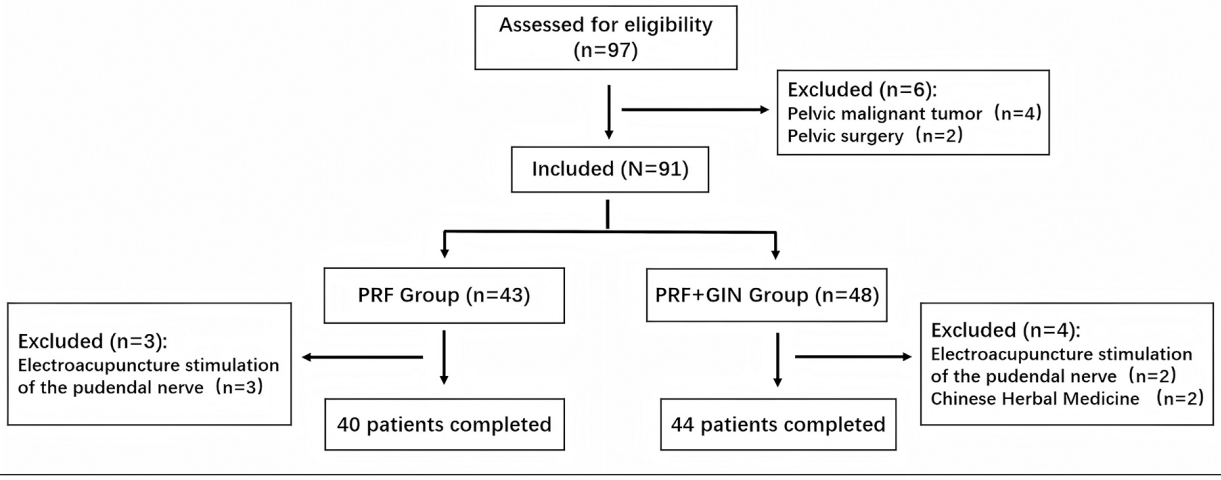


Fig. 3. *Flowchart of patient selection. PRF: pulsed radiofrequency, PRF+GIN: pulsed radiofrequency combined with ganglion impar neurolysis.*

posttreatment, respectively (Fig. 7). Both groups had statistically significant reductions at posttreatment months 3 and 6 ($P < 0.05$); the PRF+GIN Group had statistically significantly lower scores ($P < 0.05$).

PCS Scores

In the PRF Group, mean PCS scores decreased from 33.74 (10.07) at pretreatment to 17.42 (9.95) and 19.52 (9.92) at posttreatment months 3 and 6. In the PRF+GIN Group, scores decreased from 36.75 (9.87) at pretreatment to 13.47 (7.44) and 12.16 (6.16) at 3 and 6 months posttreatment, respectively (Fig. 8). Both groups had significant reductions ($P < 0.05$); the PRF+GIN Group had a statistically significantly lower score at posttreatment month 6 ($P < 0.05$).

The above measurement data are summarized in Table 3 and are in the Supplemental Material.

Adverse Reactions

Eight patients in each group experienced mild pain at the puncture site; these all resolved within 2 weeks. One patient in the PRF Group developed an acute urinary retention at posttreatment hour 2; this was managed with catheterization and resolved without further complications. No serious adverse reactions, including infection, nerve injury, or lower limb dysfunction, were reported in either group.

Discussion

Our retrospective study examined 84 women patients with PN: 40 received pudendal nerve PRF monotherapy, and 44 received pudendal nerve PRF+GIN. At posttreatment month 6,the PRF+GIN Group had greater improvements in pain intensity, anxiety, depression, and pain catastrophizing, with no severe complications observed in either group.

Unlike prior studies, our research assessed pain using both the VAS and the OIP. While the VAS quantifies pain intensity, it lacks sensitivity to pain distribution and frequency. By integrating the VAS and the OIP, our study provides a more holistic evaluation of patients' pain conditions.

The sympathetic nervous system plays a key role in neuropathic pain, including PN (17). Chronic inflammation post nerve injury sensitizes sensory neurons, increasing sympathetic excitability and exacerbating ectopic discharges (18). Sympathetic nerve modulation (19), such as GIN, is effective in managing chronic pain conditions like endometriosis, Complex Regional Pain Syndrome, and postherpetic neuralgia (11-13).

Due to a ganglion block's limited efficacy duration, it is insufficient to meet the therapeutic needs of patients with long-term chronic pain. Ganglion impar neurolysis can provide more durable pain relief. After reviewing basic research and clinical data (20-24), our study found that ganglion neurolysis is superior in pain relief degree and duration, and thus ganglion neurolysis was chosen over a simple block. Our findings indicate that GIN enhances the short-term efficacy of PRF treatment for PN by reducing sympathetic activity and disrupting the "sympathetic-sensory" feedback loop. It also improves blood circulation, promotes visceral recovery, and reduces pelvic floor tension (25), alleviating perineal pain.

Table 1. *Patient demographics.*

Variables	PRF Group	PRF+GIN Group	*P* Value
Number, n	40	44	
Age, mean ± SD	52.5 ± 17.2	55.2 ± 14.8	0.57
Body mass index (kg/m^2), mean ± SD	23.6 ± 5.2	23.2 ± 3.8	0.69
Pain duration n (%)			
< one year	7 (17.5%)	9 (20.4%)	0.78
One-3 years	24 (60.0%)	23 (52.3%)	
> 3 years	9 (22.5%)	12 (27.3%)	
Pain location n (%) Pain site			
Dorsal nerve of the clitoris	6 (15.0%)	7 (15.9%)	0.91
Perineal nerve	33 (82.5%)	39 (88.6%)	0.42
Inferior rectal nerve	13 (32.5%)	14 (31.8%)	0.95
Abdomen	7 (17.5%)	9 (20.5%)	0.73
Lumbosacral	8 (20.0%)	8 (18.2%)	0.83
Lower limbs	9 (22.5%)	9 (18.2%)	0.82
Unilateral/ Bilateral n (%)			
Unilateral	7 (17.5%)	10 (22.7%)	0.55
Bilateral	33 (82.5%)	34 (77.3%)	
Pain characteristics n (%)			
Burning	24 (60.0%)	29 (65.9%)	0.58
Stabbing	19 (47.5%)	18 (40.9%)	0.54
Throbbing	10 (25.0%)	14 (31.8%)	0.49
Aching	8 (20.0%)	10 (22.7%)	0.76
Splitting	14 (35.0%)	15 (34.1%)	0.93
Diagnostic pudendal nerve block outcomes n (%) Diagnostic Block of the Pudendal Nerve			
50% ≤ degree of mitigation <60%	4 (10.0%)	4 (9.1%)	0.99
60% ≤ degree of mitigation < 70%	3 (7.5%)	3 (6.8%)	
70% ≤ degree of mitigation < 80%	7 (17.5%)	6 (13.6%)	
80% ≤ degree of mitigation < 90%	10 (25.0%)	12 (27.3%)	
≥ 90% degree of mitigation	16 (40.0%)	19 (43.2%)	

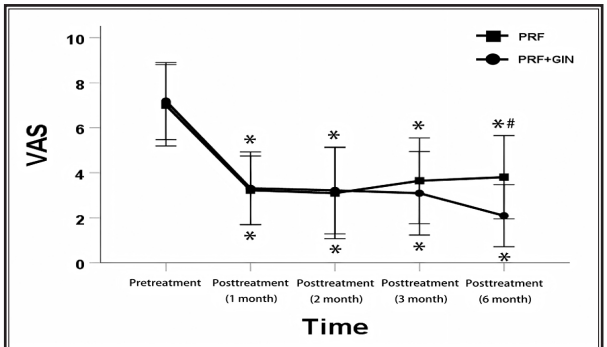


Fig. 4. *The posttreatment Visual Analog Scale scores of both were lower than those at pretreatment. At 6 months posttreatment, the PRF+GIN Group had lower scores than the PRF group. *P < 0.05, within groups compared with pretreatment #P < 0.05, PRF Group vs PRF+GIN Group.*

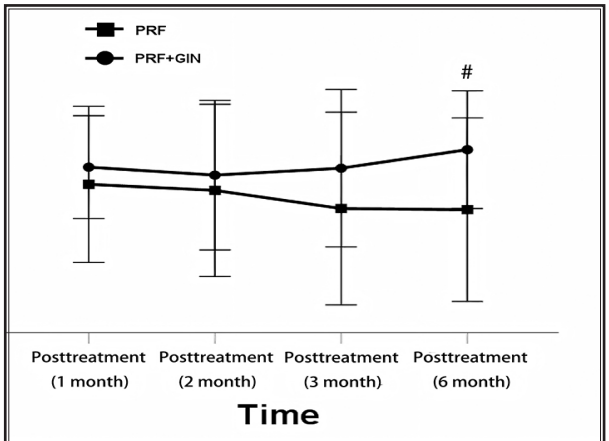


Fig. 5. *At 6 months posttreatment, the Overall Improvement in Pain value in the PRF+GIN Group was higher than that in the PRF Group. #P < 0.05, PRF Group vs PRF+GIN Group.*

The proportion of pain improvement in our study was statistically significantly higher than in other studies. For example, at posttreatment month 3, both groups in our study achieved efficacy rates exceeding 70%, while other studies reported rates of approximately 50% (8,9).This difference may be attributed to the following 2 factors: 1) in sensory testing, other researchers generally described abnormal sensations covering the pain area broadly. In our study, some patients exhibited extensive pain distribution involving 2 or more branches of the pudendal nerve, making complete coverage during treatment challenging. Target selection may influence the efficaciousness of PRF, thus, our study targeted the most severely affected or frequently painful areas during sensory testing. The nerves supplying these areas are theoretically the most affected pudendal nerve branches, leading to more pronounced treatment effects. 2) Abnormal sensations in the target area were confirmed only when sensory testing voltage was ≤ 0.3 V, ensuring the needle was

Table 2. *Treatment therapeutic effects.*

Variables	PRF Group	PRF+GIN Group	*P* Value
Effective Rate			
One month posttreatment	82.5%	86.4%	0.46
2 months posttreatment	80.0%	81.8%	0.72
3 months posttreatment	72.5%	79.5%	0.25
6 months posttreatment	65.0%	75.0%	0.12
Excellent Rate			
One month posttreatment	42.5%	47.7%	0.44
2 months posttreatment	37.5%	47.7%	0.14
3 months posttreatment	37.5%	52.3%	0.04*
6 months posttreatment	32.5%	56.8%	< 0.001*

Compared with PRF Group, *$P < 0.05$

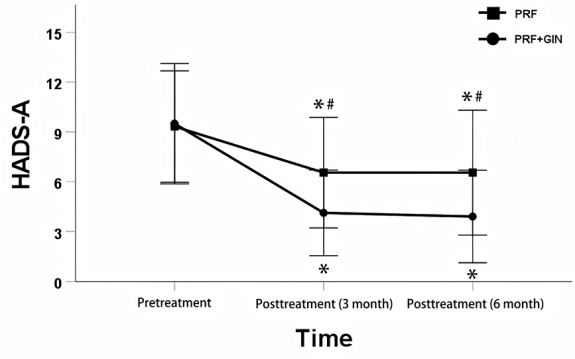


Fig. 6. *The posttreatment HADS-A scores of both groups were lower than the pretreatment scores; the PRF+GIN Group had lower scores than the PRF Group. *$P < 0.05$, within groups compared with pretreatment. #$P < 0.05$, PRF Group vs PRF+GIN Group.*

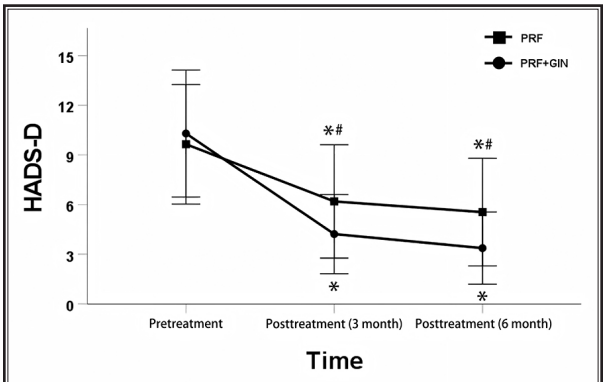


Fig. 7. *The posttreatment HADS-D scores of both groups were lower than the preoperative scores; the PRF+GIN Group had lower scores than the PRF Group. *$P < 0.05$, within groups compared with pretreatment. #$P < 0.05$, PRF Group vs PRF+GIN Group.*

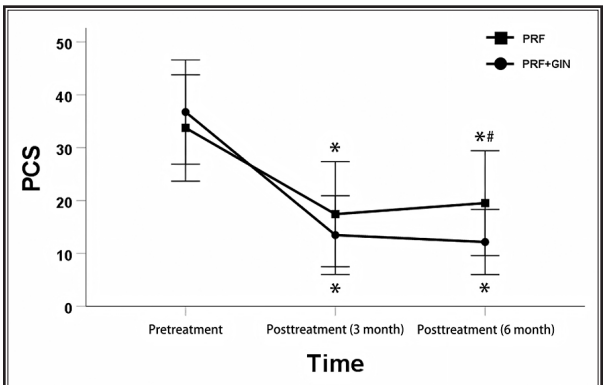


Fig. 8. *The posttreatment Pain Catastrophizing Scale scores of both groups were lower than the pretreatment scores. At 6 months posttreatment, the PRF+GIN Group had lower scores than the PRF Group. *$P < 0.05$, within groups compared with pretreatment. #$P < 0.05$, PRF Group vs PRF+GIN Group.*

positioned close enough to the nerve for optimal exposure to the PRF field. In contrast, other researchers typically set the sensory testing voltage at 0.3 V—0.5 V, which may have affected the PRF's efficacy.

Treatment efficacy in both groups gradually declined over time, likely due to the complex etiology of pudendal neuralgia, which limits some patients' sensitivity to PRF and GIN. For instance, PN caused by nerve compression was not fully addressed by the treatment in our study, leading to a gradual decline in efficacy over time. The excellent rate in the PRF Group also decreased posttreatment, which is consistent with the observed decline in efficacy. In contrast, the excellent rate in the PRF+GIN Group increased posttreatment, indicating a trend opposite to that of efficacy. A possible explanation is that post GIN, the decoupling of sympathetic and sensory nerves requires time, leading to more pronounced effects as time progresses.

No severe adverse reactions—such as infection, nerve injury, or lower limb dysfunction—occured in our study. Puncture site pain occurred in 8 patients per group, likely due to local soft tissue injury, but they resolved quickly. One patient in the PRF Group experienced acute urinary retention at posttreatment hour 2, likely due to external urethral sphincter stimulation by PRF, which was alleviated through catheterization. These findings indicate that PRF+GIN does not increase the risk of major surgical complications.

Table 3. *Posttreatment outcome measures*.

Outcome Measure	Time Point	PRF Group (Mean [SD])	PRF+GIN Group (Mean [SD])	*P* Value (Between Groups)	*P* Value (Within Group vs Pretreatment)
VAS	Pretreatment	7.00 (1.81)	7.19 (1.71)	> 0.05	-
	One month posttreatment	3.22 (1.52)	3.31 (1.62)	> 0.05	< 0.05*
	2 months posttreatment	3.09 (2.02)	3.22 (1.92)	> 0.05	< 0.05*
	3 months posttreatment	3.65 (1.91)	3.09 (1.85)	> 0.05	< 0.05*
	6 months posttreatment	3.81 (1.85)	2.09 (1.38)	< 0.05#	< 0.05*
OIP (%)	One month posttreatment	63.54 (22.88)	68.59 (15.09)	> 0.05	-
	2 months posttreatment	61.77 (25.25)	66.25 (21.92)	> 0.05	-
	3 months posttreatment	56.45 (28.26)	68.28 (23.09)	> 0.05	-
	6 months posttreatment	56.13 (26.90)	73.72 (17.27)	< 0.05#	-
HADS-A	Pretreatment	9.32 (3.35)	9.50 (3.62)	> 0.05	-
	3 months posttreatment	6.55 (3.32)	4.12 (2.57)	< 0.05#	< 0.05*
	6 months posttreatment	6.54 (3.75)	3.91 (2.70)	< 0.05#	< 0.05*
HADS-D	Baseline	9.64 (3.61)	10.30 (3.83)	> 0.05	-
	3 months posttreatment	6.20 (3.42)	4.22 (2.40)	< 0.05#	< 0.05*
	6 months posttreatment	5.55 (3.25)	3.37 (2.18)	< 0.05#	< 0.05*
PCS	Baseline	33.74 (10.07)	36.75 (9.87)	> 0.05	-
	3 months posttreatment	17.42 (9.95)	13.47 (7.44)	> 0.05	< 0.05*
	6 months posttreatment	19.52 (9.92)	12.16 (6.16)	< 0.05#	< 0.05*

* Indicates a statistically significant difference ($P < 0.05$) within the group compared to baseline. # Indicates a statistically significant difference ($P < 0.05$) between the PRF and PRF+GIN groups. VAS, Visual Analog Scale; OIP, Overall Improvement in Pain; HADS-A, Hospital Anxiety and Depression Scale-Anxiety; HADS-D, Hospital Anxiety and Depression Scale-Depression; PCS, Pain Catastrophizing Scale.

Limitations

Our study has several limitations. First, as a retrospective analysis with a small sample size. The synergistic effect of GIN and pudendal nerve PRF requires large-scale, prospective, randomized controlled trials. Second, the 6-month follow-up period limits the assessment of long-term efficacy. Additionally, the study's exclusive focus on women patients, due to the higher prevalence of PN in women, restricts the generalizability of the findings. Future research should include men in order to evaluate potential gender differences in treatment outcomes.

Conclusion

In conclusion, GIN may enhance the short-term efficacy of pudendal nerve PRF for managing PN, improving both pain relief and psychological well-being.

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